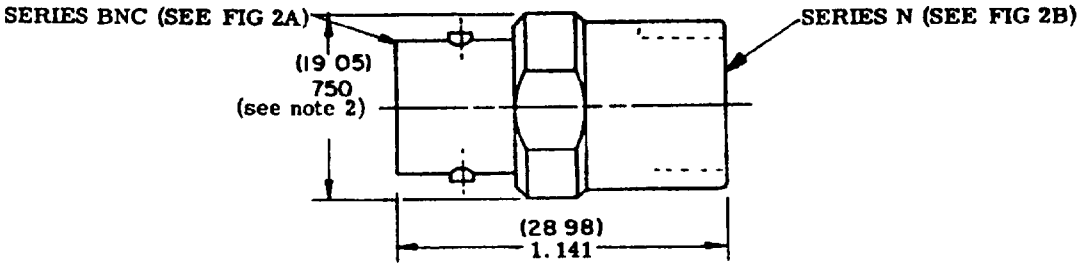
MILITARY SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (SERIES BNC TO SERIES N), CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapter described herein shall consist of this document and the latest issue of Specification MIL-A-55339.



NOTES:

1. Dimensions are in inches
2. This dimension is the largest overall diameter of the adapter
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm

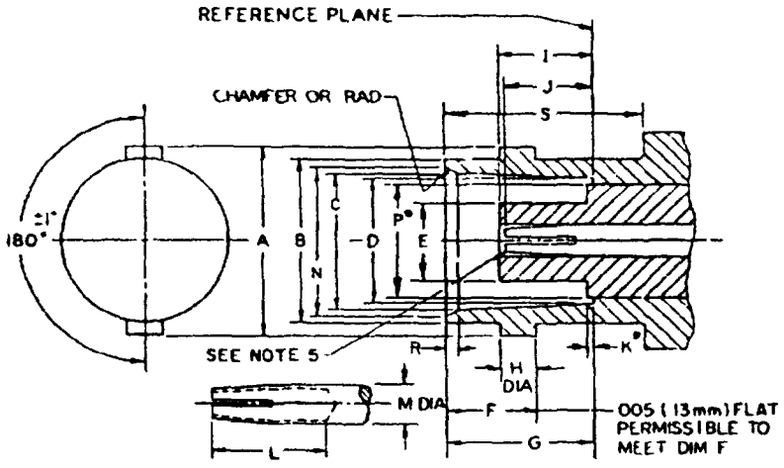
FIGURE 1 General configuration

DESIGN AND CONSTRUCTION

General configuration	See figure 1
Impedance	50 ohms, nom
Working voltage	
Sea level	500V rms
70,000 feet	125V rms
Frequency range	0 to 4,000 MHz
Temperature range	-65° to +165° C

(A) denotes changes

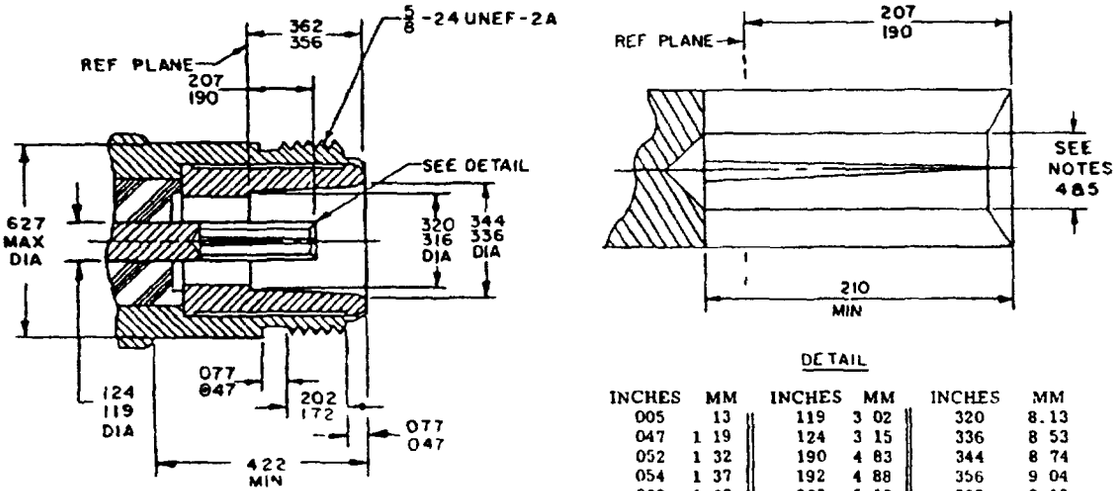
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.432(10.97)	.436(11.07)
B	.378 (9.60)	.382 (9.70)
C	.327 (8.31)	.333 (8.46)
D	.319 (8.10)	.321 (8.15)
E	- - - -	.186 (4.72)
F	.204 (5.18)	.208 (5.28)
G	.327 (8.31)	.335 (8.51)
H	.075 (1.91)	.081 (2.06)
I	- - - -	.208 (5.28)
J	.190 (4.83)	.206 (5.23)
K*	- - - -	.006 (.15)
L	.195 (4.95)	- - - -
M	.081 (2.06)	.087 (2.21)
N	.346 (8.79)	.356 (9.04)
P*	- - - -	.256 (6.50)
R	.015 (.38)	.030 (.76)
S	.414(10.52)	- - - -

* P dimension applies to that portion (if applicable) of dielectric which extends beyond reference plane by dimension K*.

FIGURE 2A Series BNC (see note 3)



DETAIL

INCHES	MM	INCHES	MM	INCHES	MM
005	13	119	3 02	320	8.13
047	1 19	124	3 15	336	8 53
052	1 32	190	4 83	344	8 74
054	1 37	192	4 88	356	9 04
063	1 60	202	5 13	362	9 19
066	1 68	207	5 26	422	10 72
077	1.96	210	5 33	627	15 93
100	2 54	.316	8 03		

FIGURE 2B. Series N (see note 4)

NOTES
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. For series BNC only, concave depression .100 x 005 in. deep between studs is permissible.
4. For series N only, slitting of center contact is optional
5. ID to meet VSWR and contact resistance when mated with 054/ 052 dia pin (series BNC), or .066/ 063 dia pin (series N)

FIGURE 2 Mating dimensions

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PERFORMANCE (installation torque of 6 to 10 in lb, series N only)

Dimensions - See figures 1 and 2.

Center contact retention
Axial force - 6 lb, min.
Torque - 4 in oz, min

Force to engage and disengage:
See table I.

Coupling proof torque: 15 in. lb, max
(series N only)

Mating characteristics·
Center contact - See table I.
Outer contact - Not applicable

Permeability· Less than 2.0

Seal (hermetic, pressurized, and
weatherproof)· Not applicable

Insulation resistance Method 302 of
MIL-STD-202, test condition B.
5,000 megohms, min.

VSWR: 1.30, max, at 500 to 4,000 MHz.

RF leakage (total)· -55 dB, min, at
2 to 3 GHz.

RF insertion loss: .2 dB, max, at 3 GHz
(.12 $\sqrt{F(GHz)}$ dB max tested at 3 GHz)

Durability· 500 cycles, min, at 12 c/m

Dielectric withstanding voltage· Method 301
of MIL-STD-202.
Test voltage - 1,500V rms, max, at
sea level

Contact resistance (in milliohms, max).

	Initial	After environment
Center contact	2 0	2 5
Outer contact	0 2	0 2

Vibration, high frequency. Method 204 of
MIL-STD-202, test condition B

Shock (specified pulse): Method 213 of
MIL-STD-202, test condition B.

(A) Thermal shock Method 107 of
MIL-STD-202, test condition B.

Moisture resistance Method 106 of
MIL-STD-202
Insulation resistance - 200 megohms, min

Corona level: 375V, min
Altitude 70,000 feet

RF high potential withstanding voltage
RF voltage - 1,000V rms
Frequency - 5 MHz

Salt spray (corrosion). Method 101 of
MIL-STD-202, test condition B.

Coupling mechanism retention force
Not applicable

(A) MARKING· As specified in MIL-A-55339
Part No M55339/01-00001.

TABLE I. Performance characteristics

Characteristic	Test value	
	Series BNC	Series N
Force to engage and disengage·		
Longitudinal force	3 lb, max	3 lb, max
(A) Torque	2-1/2 in. lb, max	10 in lb, max
Mating characteristics (center contact):		
Oversize test pin (inserted .125 in. deep)	057 in dia	074 in dia
Max test pin (insertion force 2 lb, max)	054 in dia	066 in dia
Min test pin (withdrawal force 2 oz, min)	052 in dia	063 in dia

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Navy - EC
Air Force - 11

Review activities
Army - MU, MI, EL
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities
Army - AT
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent
DSA - ES

(Project 5935-1986-1)

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a. Paragraph Number and Wording

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